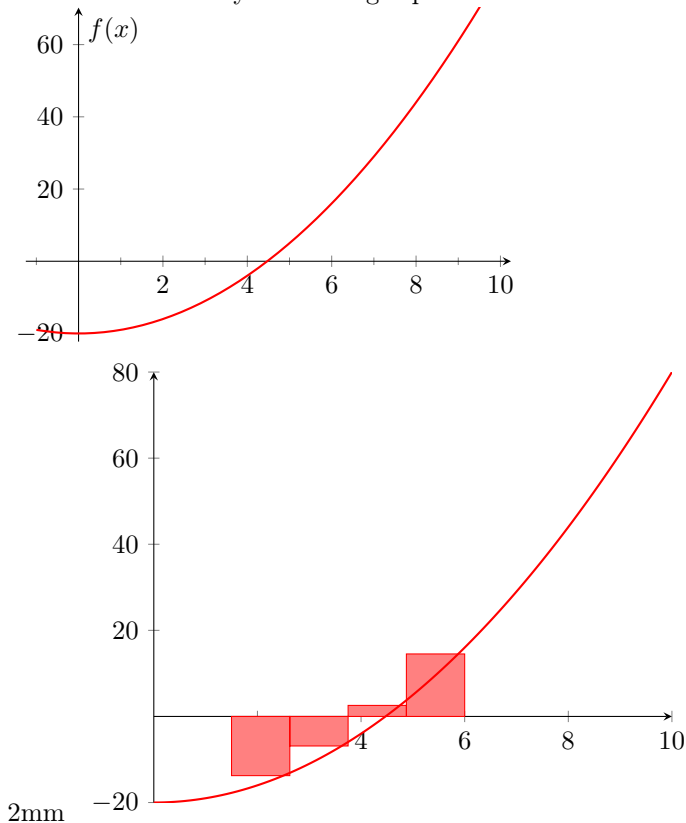


Justify your answers and show your work.

Complete the following steps for $f(x) = x^2 - 20$, interval $[1, 9]$, and $n = 4$.

1. Sketch the graph of the function on the given interval and illustrate the *right Riemann sums* using n subintervals. Clearly label the gridpoints.



2. Calculate the *right Riemann sums* for the given interval and number of subintervals, n .

Evaluate the following using the provided values.

3. $\int_0^{\pi} g(x) dx$

$$\int_0^{\pi/2} g(x) dx = 1$$

$$\int_{\pi/2}^{\pi} g(x) dx = \pi - 1$$

$$\int_{\pi}^{3\pi/2} g(x) dx = \pi + 1$$

$$\int_{3\pi/2}^{2\pi} g(x) dx = 2\pi - 1$$

4. $\int_{\pi}^0 \frac{g(x)}{\pi} dx$

5. Suppose that $g(x)$ is *even*,
find $\int_{-\pi/2}^{\pi/2} g(x) dx$.

6. Evaluate the expression $\sum_{n=0}^2 (3n + 1)$ by writing it as the sum of each of its terms.

Use geometry or symmetry to evaluate the definite integral. Recall that the area of a triangle is $A = \frac{1}{2}bh$.

7. $\int_{-2}^2 (2 - |x|) \, dx$

Use the value of the first integral I to evaluate the two given integrals. $I = \int_a^b (x - x^2) \, dx = -1$.

8. $\int_a^b (3x^2 - 3x) \, dx$

Evaluate the following.

9. Let a and b be constants.

a. $\frac{d}{dx} \int_a^x 3t^2 \, dt$

b. $\frac{d}{dx} \int_a^b f(t) \, dt$

10. $\int_0^2 (x^2 + 1) \, dx$

11. $\int_0^\pi \frac{1}{2} \sin \frac{\theta}{2} d\theta.$

12. $\int \frac{1 - z^4}{z^2} dz$

13. $\int x^3 (x^4 + 4)^{15} dx$

14. $\int \frac{1}{(5x - 3)^2} dx$

15. $\int \cos x \sin^3 x \, dx$

Recall that $\sin x$ is an odd function.

16. $\int_{-5}^5 \sin^3 x \, dx$

Solve the following word problems.

17. An arch is 10 ft high and a 20 ft base. Its shape can be modeled by $y = 10 - \frac{x^2}{10}$. Find the average height of the arch above the ground. Note that the maximum height is at $x = 0$.

18. A man is walking in a straight line away from JCSU at the rate of 2 mi/hr. If he is 1 mile from JCSU when he starts walking ($t = 0$), find a function describing his distance from JCSU.